Answer Key & Teacher Notes

Session 1 — Fractions, Decimals, Percents · Session 2 — Compare and Order

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages: 8 problems, Find the Error, 2 test items. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Teaching Support Guide	Teacher page: lesson route, what to look for on each exit item, the next prompt, and a fresh check for after reteaching.
Explain Your Thinking	Optional language support for the Guided Notes "Model, Reason, Explain" page: draw it, talk it, write it.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank (English and Spanish), step frames on every problem and larger type (12 pt).

SESSION 1 · TEACHER NOTES

Fractions, Decimals, Percents

TALK PROMPTS

- How does writing an equivalent fraction with a denominator of 100 help you read the percent?
- Why is 0.5 equal to 50% and not 5%? Use the word hundredths.

PARTNER TASK

Free Throws and Phone Battery. One partner writes the fraction or decimal; the other writes the percent and explains the scaling. Switch roles with Jacket Sale: $26 off $40 (26/40 = 13/20 = 0.65 = 65% discount).

WATCH FOR

- Reading 0.08 as 80% or 0.7 as 7% (misplacing the tenths and hundredths digits).
- Writing the part as the percent without scaling to 100 ($26 off $40 called a "26%" discount).
- Using the wrong whole: comparing 8 red to 12 blue beads instead of 8 to the total of 20.

SESSION 2 · TEACHER NOTES

Compare and Order

TALK PROMPTS

- Which form did your group choose for comparing, and what made that form easier to use?
- How can you tell 4/5 lands between 0.7 and 86% without dividing?

PARTNER TASK

Quiz Scores. One partner writes 0.76, 18/25, and 79% as percents (76%, 72%, 79%); the other orders them least to greatest and checks. Switch roles with Rainfall: 0.12, 3/20, and 9%.

WATCH FOR

- Comparing digits instead of values (0.4 < 39% "because 4 < 39").
- Treating a longer decimal as larger, or reading 0.6 as 6%.
- Missing ties: 0.36 and 9/25 are both 36% and should sit side by side.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 · Form A	6.AT.4	1	2	1	1	2	2	1	3	3	1	1
S1 · Form B	6.AT.4	1	2	1	1	2	2	1	3	3	1	1
S2 · Form A	6.AT.4	1	2	1	1	2	2	2	3	3	1	1
S2 · Form B	6.AT.4	1	2	1	1	2	2	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

percent: A ratio that compares a number to **100**.

equivalent: Names the **same** amount in a different way.

hundredths: Parts out of 100; the **second** place after the point.

- Fraction to percent: make an equivalent fraction with a denominator of **100**.
- Decimal to percent: read the decimal in **hundredths**, so 0.4 = 0.40 = 40%.
- Percent to fraction: write the percent over 100, then **simplify**.

Example: Fraction, part over whole: **7/20**. 20 × **5** = 100, so 7/20 = 35/100. 35/100 = **0.35** as a decimal. Answer: 7/20 = 0.35 = **35%**

YOU TRY

1. a. 60% b. 8% c. 70%

2. 0.92 and 92/100 = 23/25

EXIT TICKET

1. a. 9% b. 65%

2. a. 18/25 b. 0.72 c. 72% (18/25 = 72/100)

3. **C** (4/100)

PRACTICE FORM A

1. a. 60% b. 45% c. 8% d. 12% e. 72% f. 25%

2. Bar: 40%, 60%, 80% (10 votes = 20%, so 1 vote = 2%). Hamster: 16/50 = 8/25 = 0.32 = 32%. Fish: 24/50 = 12/25 = 0.48 = 48%. Turtle: 50 − 40 = 10 votes, 10/50 = 1/5 = 0.2 = 20%.

3. a. 12/16 = 3/4 b. 75%

4. a. 60% b. 40%

5. 26/40 = 13/20 = 65/100 = 0.65 = 65% discount

6. a. 8/20 = 40/100 = 40% b. 0.4

7. a. 45% b. 0.45 c. 45/100 = 9/20

8. a. A 45%, B 44%, C 46% b. B, A, C

9. The 8 is in the hundredths place, so 0.08 = 8/100 = 8%. 80% would be 0.80 (8 tenths).

10. **C** (80%)

11. **B** (0.25)

Self-check word **PERCENT**

PRACTICE FORM B

1. a. 90% b. 55% c. 4% d. 24% e. 38% f. 75%

2. Bar: 50%, 75% (10 votes = 25%, so 2 votes = 5%). Museum: 14/40 = 7/20 = 0.35 = 35%. Zoo: 18/40 = 9/20 = 0.45 = 45%. Nature center: 40 − 32 = 8 votes, 8/40 = 1/5 = 0.2 = 20%.

3. a. 15/25 = 3/5 b. 60%

4. a. 35% b. 65%

5. 22/40 = 11/20 = 55/100 = 0.55 = 55% discount

6. a. 6/20 = 30/100 = 30% b. 0.3

7. a. 64% b. 0.64 c. 64/100 = 16/25

8. a. A 62%, B 60%, C 61% b. B, C, A

9. The 7 is in the tenths place: 0.7 = 0.70 = 70/100 = 70%. 7% would be 0.07.

10. **C** (30%)

11. **B** (0.75)

Self-check word **DECIMAL**

GUIDED NOTES

common form: Every value written the **same** way before you compare.

benchmark: An easy number you know well, used to **compare** other numbers.

order: To list numbers from **least** to greatest, or greatest to least.

- To compare a fraction, a decimal and a percent, first write them all as **percents**.
- After you order the percents, write the list with the **original** numbers.

Example: 0.7 = 0.70 = **70%** 3/5 = 60/100 = **60%**
Compare: 60% < 65% < 70% Answer: Least to greatest: **3/5**, 65%, 0.7

YOU TRY

1. a. > b. = c. <

2. 62%, 60%, 61%, so 3/5, 61%, 0.62

EXIT TICKET

1. a. > b. < c. =

2. 35%, 37.5%, 36%, so 0.35, 36%, 3/8

3. **B** (3/4)

PRACTICE FORM A

1. a. > b. < c. = d. < e. < f. >

2. a. Red 90%, Blue 80%, Green 86%, Gold 70% b. Gold, Blue, Green, Red

3. a. 76%, 72%, 79% b. 18/25

4. a. 12%, 15%, 9% b. 3/20

5. No. 0.4 = 0.40 = 40%, and 40% > 39%, so 0.4 is greater.

6. Heroes 35%, villains 40%, sidekicks 25% (total 100%); villains are the largest group.

7. a. 36%, 36%, 34% b. 34%, then 0.36 and 9/25 (tie) c. 0.36 and 9/25 tie at 36%

8. a. Ana 62%, Ben 62.5%, Cal 61%, Dee 60% b. Dee, Cal, Ana, Ben

9. Owen compared digits, not values. 0.55 = 55%, 3/5 = 60/100 = 60%, and 61%. 3/5 is not the smallest. Order: 0.55, 3/5, 61% (55% < 60% < 61%).

10. **A** (0.9)

11. **C** (1/4)

Self-check word **VALUES**

PRACTICE FORM B

1. a. > b. < c. = d. < e. < f. >

2. a. Maple 85%, Oak 90%, Pine 83%, Cedar 75% b. Cedar, Pine, Maple, Oak

3. a. 88%, 84%, 86% b. 21/25

4. a. 14%, 16%, 11% b. 4/25

5. No. 0.3 = 0.30 = 30%, and 30% > 29%, so 0.3 is greater.

6. Animals 45%, stars 30%, hearts 25% (total 100%); animals are the largest group.

7. a. 48%, 48%, 46% b. 46%, then 0.48 and 12/25 (tie) c. 0.48 and 12/25 tie at 48%

8. a. Eli 84%, Fay 87.5%, Gus 86%, Hana 80% b. Hana, Eli, Gus, Fay

9. Nadia compared digits, not values. 0.75 = 75%, 4/5 = 80/100 = 80%, and 82%. 4/5 is not the smallest. Order: 0.75, 4/5, 82% (75% < 80% < 82%).

10. **A** (0.4)

11. **C** (1/2)

Self-check word **SHARES**

SHEET 1 · FRACTIONS, DECIMALS, PERCENTS

1. a. 35/100 = 7/20 b. 0.35 c. 35%
2. 1/4 = 0.25 = 25%; 7/10 = 0.7 = 70%; 2/25 = 0.08 = 8%; 13/20 = 0.65 = 65%; 3/5 = 0.6 = 60%
3. 11/25 = 44/100 = 0.44 = 44%
4. She just wrote the digits 3 and 4 side by side; that is not the same value. 3/4 = 75/100 = 0.75 = 75%.

SHEET 2 · COMPARE AND ORDER

1. a. > b. < c. <
2. a. 7/10, 0.72, 75% b. 0.2, 22%, 1/4 c. 6/5, 125%, 1.3
3. A = 55%, B = 60%, C = 52%. Order: Room C, Room A, Room B (Room B greatest).
4. 3/5 = 60% and 0.7 = 70%, so any value between 60% and 70%. Sample: 13/20 = 0.65 = 65%.
5. Comparing single digits does not compare values; change to one form. 3/8 = 0.375 = 37.5% and 0.4 = 40%, so 3/8 < 0.4.
6. **C** (0.35, 38%, 2/5)
7. **C** (64%)

SESSION 1 · FRACTIONS, DECIMALS, PERCENTS

Percent row: 0%, 12.5%, 25%, 37.5%, 50%, 62.5%, 75%, 87.5%, 100%. a. 1 slice = 100% ÷ 8 = 12.5%; 3 slices = 3 × 12.5% = 37.5%. b. 3 ÷ 8 = 0.375. c. 3/8 means 3 of 8 equal parts. On the bar, 3 parts end at 37.5%. 0.375 = 37.5 hundredths = 37.5%, so 3/8 = 0.375 = 37.5%.

Challenge Each slice is 12.5%, so 3 slices are 3 × 12.5% = 37.5%, not 38%. Writing the digits 3 and 8 side by side does not keep the value: 38% = 38/100, but 3/8 = 37.5/100.

SESSION 2 · COMPARE AND ORDER

a. 0.45 = 45%; 2/5 = 40/100 = 40%; 48% = 48%. b. Class B (2/5), Class A (0.45), Class C (48%). c. With all three as percents, 40% < 45% < 48%. On the line, 2/5 sits at 40%, 0.45 at 45%, and 48% is farthest right.

Challenge Leo compared single digits, not values. 2/5 = 40/100 = 40% and 0.45 = 45%. 40% < 45%, so 2/5 is less than 0.45.

Score each task 0–3: 1 for a model that fits the numbers, 1 for correct math with units, 1 for an explanation that matches the model.